

**ECON 101—Introduction to Microeconomics
Spring 2023**

Final

April 30, 2023

Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15

Bonus question: What is the name of Prof. Alejandro's cat? _____

MCQ (Problems 1 and 2)	Problem 3.1	Problem 3.2	Bonus	Total

Version	I	II	III	IV
Question				
1	(c)	(d)	(a)	(c)
2	(a)	(c)	(b)	(c)
3	(c)	(a)	(b)	(b)
4	(c)	(c)	(c)	(b)
5	(a)	(b)	(d)	(d)
6	(c)	(d)	(a)	(b)
7	(b)	(d)	(d)	(d)
8	(c)	(a)	(c)	(c)
9	(a)	(c)	(c)	(d)
10	(c)	(b)	(c)	(d)
11	(d)	(a)	(c)	(b)
12	(a)	(b)	(c)	(d)
13	(a)	(c)	(b)	(a)
14	(d)	(d)	(c)	(b)
15	(c)	(b)	(a)	(a)

Problem 3: True/False questions

For the next two statements, write whether it is true or false, and justify your answer. Your argument should be short, answer it in the space provided.

1. Consider the market for cars. All firms have the same production technology, and we know that the long-run average cost curve has a minimum at a cost of 4,600,000 Tenge per car. Therefore, we can never observe a market equilibrium in which the equilibrium price is below 4,600,000 Tenge per car.

FALSE. There can be a short-run equilibrium with a price below 4,600,000 Tenge because the firms may not want to shut down even if they are making negative profits, as long as they can cover their variable costs.

2. Consider a firm that produces computers. Suppose that the market price of a computer is $\bar{p}$, both in the short-run and the long-run. If in the short-run this firm decides to shut-down, then it must be that in the long-run it decides to exit the market.

FALSE. It is possible for the firm to decide to shutdown in the short run, but if it were able to choose another level of the fixed input then it could produce more efficiently and stay in the market.

A more in-depth explanation: The decision to shut down is taken in the short-run with respect to the short-run average variable cost curve, i.e., if the firm decides to shut down if $\bar{p}$ is less than the shutdown price. However, the decision to enter or exit the market is taken with respect to the long run average cost curve, i.e., if $\bar{p}$ is less than the break-even price implied by the long run average cost curve. Remember that the long-run average cost curve envelopes from below any short-run average total cost curve. The long run cost curves are showing the optimal way for the firm to produce, in terms of minimizing its costs while being able to change any of its inputs. So a firm may decide in the short-run either to shutdown, or if it is still possible, to alter its production plan by altering its input usage in the long run to achieve a lower cost.

Write here anything you want me to consider about Problem 1 for partial credit.

Write here anything you want me to consider about Problem 2 for partial credit.